

Blast Optimization and Vibration Control at a Multi-Seam Coal Mine, Brazil

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Abstract

The B3 mine is a multiple coal seam strip mining in operation in Rio Grande do Sul, the southernmost state in Brazil. A traditional truck & excavator operation is used to remove the overburden and selectively mine coal seams, ranging in thickness from 30 cm (1 ft) to 2.5 m (8.2 ft). Blasting is required in order to excavate part of the sedimentary rock overburden where the rock is too hard for mechanical excavation, and to mine the coal seams, selectively.

The blast design is tailored for each coal seam individually or can be combined for two or more coal seams, depending on the thickness of the seam and on the intermediate waste rock. In order to prevent the mixing of coal and waste (coal dilution), powder factors normally used in blast design are very low, as the main blasting goal is to induce more fractures in the rock, reducing its strength to facilitate excavation.

Extensive vibration monitoring was conducted at the mine attempting to improve blasting results. Attenuation equations were also obtained to estimate the effects of changes in blast patterns, the hole to hole detonation sequence and the use of different combinations of delay timing periods.

For the blasting detonation sequence, a typical “serpentine” tying method was used in hole to hole initiation. This method guarantees a low charge weight per delay as only one hole is detonated per delay period. “Parallel” and “diagonal” tying methods were then applied and different delay period combinations used. These changes resulted in improvements in the blast performance such as better rock fragmentation, increased excavator production and lower vibration levels, even when the charge weight per delay was increased.

Due to the improved blast results, costs were reduced because it was possible to increase burden and spacing without increasing vibration levels. Costs were also reduced as the amount of explosives, accessories and also the total meters drilled were diminished.

Introduction

Copelmi Mineração operates the B3 mine, a multiple coal seam strip mining site in Rio Grande do Sul, the most southern state in Brazil. A traditional truck & excavator operation is used to remove the overburden and selectively mine coal seams.

The coal deposits in the region are characterized by the presence of many coal seams intercalated by waste material, typically siltstone and sandstone. Mined coal thickness ranges from 30 centimeters (1 foot) to 2.5 meters (8.2 feet) with interburdens ranging from a few centimeters to more than 12 meters (40 feet). Overburden varies from 5 meters (16 feet) to over 40 meters (130 feet). Figure 1 shows the profile for the B3 mine coal deposit.

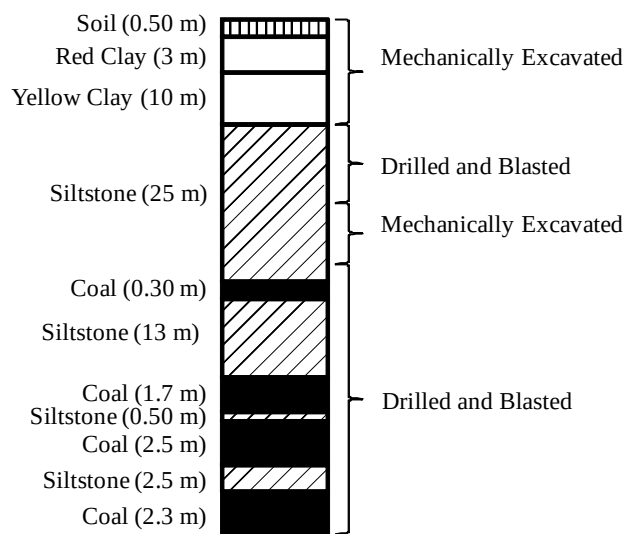


Figure 1. Coal deposit profile.

Using strip mining it is possible to mine more coal seams in the deposits depending on the overburden ratio. A typical coal processing includes the multiple seams altogether, regardless to the individual properties such as ash percentage, sulfur content and calorific values. In the early 90s coal processing used to be carried out individually to obtain better results in recovering percentage and to increase production, as each seam has different properties. Nowadays, selectivity has been introduced and, besides mining and processing the individual coal seams, some of the seams are divided in parts, according to quality. After so, these parts are individually mined and processed.

In 2010 the blast optimization study started with the goal of improving rock fragmentation and excavator production rates, reducing vibrations and noise in the surrounding areas and determining the best blast pattern for each blasted material. Vibration and noise levels are always considered an issue. In this case, the mine site is located very close to a small town. The nearest home is located 300 meters (970 feet) from blasting operations and in the near future this distance will be 100 meters (320 feet). Determining the best blast design, the one that will produce the lowest vibration levels, is a key factor for the future mining.

Blast design characterization was conducted during the initial phase of the blast optimization process. Blast plans were analyzed and blast oversight was conducted. Vibration monitoring was carried out and attenuation equations were obtained to compare changes applied to blast design. Later, changes in blast initiation sequence were proposed. Different tying methods were used and varied delay periods were tested using data obtained from vibration monitoring. After all, changes to burden and spacing as well as blasthole lengths were applied and the results were compared to find the best blast design for each material.

Blast Design Overview

The coal seams and waste material are mainly composed of siltstones and do not require great amounts of energy to be disaggregated and excavated. For that reason, powder factors used in blast designs are very low, resulting in tight and confined blasts with little rock movement. A minimum amount of energy is used just to induce fractures to the coal and siltstones in order to facilitate the excavation. Another reason for the low powder factor and little movement of the material is to prevent coal dilution which may occur when too much energy is used and there is a vertical or horizontal movement. Powder factors used range from 70 g/m³ (0.118 lbs/yd³) to 280 g/m³ (0.472 lbs/yd³).

Selective mining requires the seams to be blasted and excavated individually, to avoid dilution. Thus, blasting has to be done separately for each coal seam. To prevent oversized fragments of coal, a layer of siltstone is left on top of the seam and drilled and blasted along. This causes the column charge to be moved up, filling the entire coal thickness and resulting in better fragmentation as more energy is released in the seam. If no siltstone is left on top, the stemming column is placed in the upper portion of the seam, resulting in poor fragmentation in the areas around the stemming column.

The materials composing the overburden do not always require the use of explosives before the excavation process as some parts of siltstone are soft enough and excavators can dig without losing productivity. When productivity falls, the material is drilled and blasted. The coal seams are blasted, regardless the possibility of being mechanically excavated, because with explosives it is possible to obtain better fragmentation at a lower cost.

Blastholes are drilled 76 mm (3") in diameter, lengths varies according to coal or siltstone thickness, burden and spacing varies depending on each material and holes are drilled in a staggered pattern. Explosives used are emulsion cartridges and bulk ANFO. Emulsion cartridges are used as primers to initiate the ANFO column (Munaretti, 2002). Shock tubes with and without delay timing are used to initiate the explosive columns and on surface connections. Delay periods used have low accuracy, between 23 % and 46 % scatter, making maximum number of holes per delay difficult to predict. More precise delays are available in the Brazilian market but they cost 3 to 4 times more.

Initiation Sequence

It is well known that the determination of the right or the best hole to hole initiation sequence is one of the fundamental keys for a well performed blast. It is the detonation sequence that determines how the energy of the explosives will be released during detonation and be used in the movement and

fragmentation of the material. The more efficient the detonation sequence is and, by consequence, the more efficient the blast, better is the level of energy released during detonation used in the movement and fragmentation of the rock, and lower is the energy left to be transferred to the rock mass surrounding or to the atmosphere in the form of vibration and noise (Dowding, 1985).

The traditional initiation sequence used for many years was the “serpentine” initiation sequence. This tying method consists in connect hole to hole in sequence, or group of holes together, such as each hole is detonated individually throughout the entire detonation sequence. This method limits to one the maximum number of holes per delay (or to the number of holes in a group of holes tied together) and has a low charge weight per delay. A lower degree of freedom is also typical. Examples of the serpentine initiation sequence are shown in figure 2(a) and (b).

With this initiation sequence, there is none or little horizontal movement towards the free face and vertical movement is evidenced. This type of shot is considered to have high confinement and only one degree of freedom, the surface of the shot. In this method, only one type of delay period is normally used.

One of the disadvantages of this initiation sequence is that it limits the hole lengths. It was observed that when drilling deep holes, over 4 meters (13 feet), the bottom part of the shot could not be dug and had to be drilled and blasted again. Also, due to confinement, it produced high ground vibration and air overpressure (noise), flyrock and the total time of shot was long depending on the number of holes drilled and such results can be annoying to neighbors.

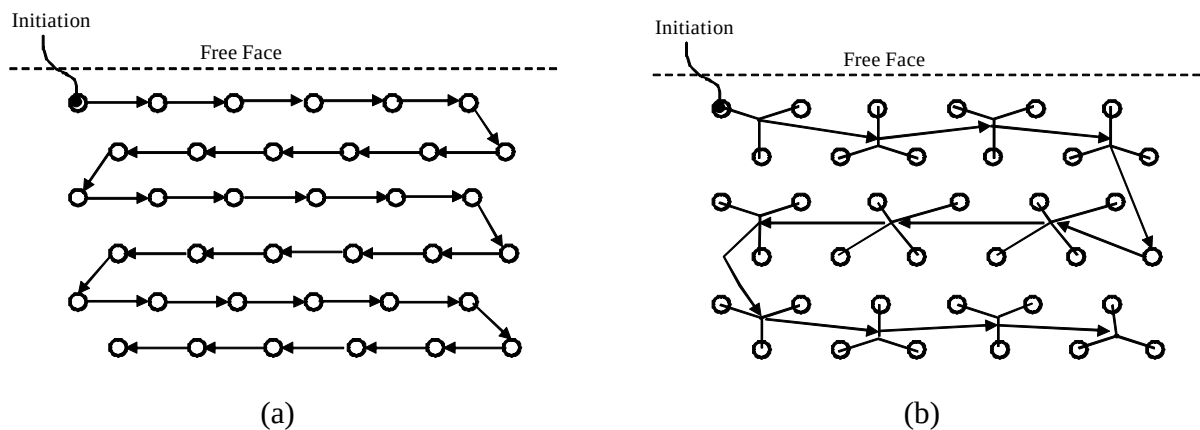


Figure 2. Serpentine initiation sequence, (a) one hole per delay and (b) a group of holes per delay.

After blast plans analysis and conducting blasting oversight, a “parallel” initiation sequence was proposed. This method consists in connecting hole to hole in parallel lines initiation towards the same direction and using a control or base line to initiate each individual line. Figure 3(a) shows an example of the parallel initiation sequence. Delay periods can be varied using one timing period for hole to hole initiation and another in between lines. This method is considered to have two degrees of freedom, because besides the surface of the blast site the bench free face is used and some horizontal movement is noticed even when using low powder factors.

This method proved to be more efficient, producing better fragmentation, eliminating the need of drilling and shooting again, reducing ground vibration and noise, flyrock and the total duration of the shot. Different delay period combinations were performed to determine the best timing combinations resulting in low ground vibrations. The improvements provided by the changes in the initiation sequence allowed changes in the blast design parameters, such as increasing burden and spacing and hole lengths.

A third initiation sequence was tested, called the “diagonal” sequence. In this method the control line is parallel to the free-face and lines are tied diagonally from the control line as shown in figure 3(b). This method has the same characteristics as the parallel method, two degrees of freedom and the possibility of using different combinations of delay periods. Both initiation sequences tested have more than one hole firing at the same time or at an 8 millisecond interval and might have a higher charge weight per delay than the serpentine method.

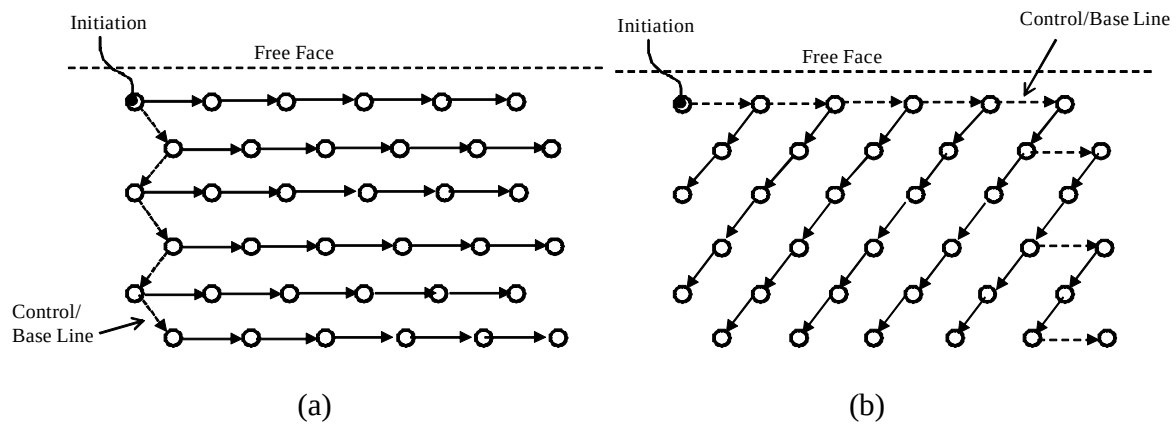


Figure 3. (a) Parallel and (b) diagonal initiation sequence.

Results

Attenuation analysis of ground motions were conducted to compare changes made in blast design, such as initiations sequence, delay period combinations and burden and spacing. Monitoring was conducted at distances from the blasts varying from 35 to 1100 meters (113 to 3500 feet) using blasting seismographs.

The ideal way of measuring the vibration and noise impact is to use many seismographs in a line for each blast in order to analyze the decay of the ground vibration intensities with distance for the same energy source. For this project, one and, sometimes, two seismographs were available. For that reason, a great number of blasts were monitored for each modification in the design in order to record ground vibrations at different distances and to be able to produce attenuation equations.

The data collected was plotted in the peak particle velocity (PPV) versus scaled distance (SD) attenuation plot. Best-fit lines through the data were obtained using a power curve of the form:

$$PPV = a \cdot SD^{-b}$$

Where, “a” is the y-intercept at SD equals to 1 and “b” the slope of the attenuation line. The “a” factor measures the intensity of the seismic energy that is transferred into the ground and is propagated away. It is a function of confinement, explosives density and energy. The slope term “-b” is related to the geology through which the seismic wave passes and is a measure of the decay of the velocity intensity with distance (Rosenhaim et al, 2005, Aimone-Martin et al, 2003).

For each initiation sequence tested and different delay period combinations used an attenuation equation was obtained and the results are presented in figure 5. In this graphic the data for all materials blasted, coal and siltstone were combined and separated according to initiation sequence and delay periods tested. Delay period combinations are represented as hole to hole versus line delays, for example 30x60 ms (milliseconds) refers to 30 ms hole to hole and 60 ms line to line (control line). “P” represents the parallel initiation sequence and “D” the diagonal.

Figure 5 shows reduction in ground motion intensities with changes in initiation sequence. Variations in peak particle velocities with the use of different delay periods are also evidenced. The switch from serpentine tying method to parallel resulted in an improved and less confined blast, shown by the progressive reductions in the y-intercept factor “k” from 4261(serpentine) to 750 (30x60 ms parallel sequence). Another reduction on vibration intensities can be seen when the diagonal sequence was tested resulting in an even lower y-intercept of 391 for the 30x30 ms combination. Attenuation velocities (slope term “-b”) are similar in all tests conducted.

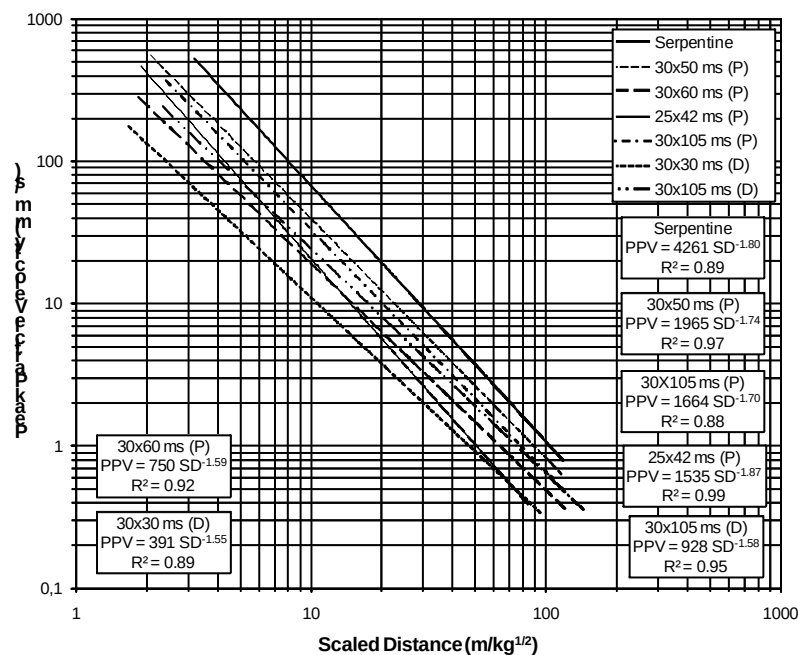


Figure 5. Peak particle velocity versus scaled distance attenuation plot comparing serpentine, parallel (P) and diagonal (D) initiation sequences and delay period combinations.

Table 1 summarizes average and maximum charge weight per delay for each test conducted. It is possible to state that with the better use of the explosive energy in the movement and fragmentation of the materials vibration levels were reduced even when raising the charge weight per delay. With the diagonal method it was possible to maximize the explosive energy use in what is most important, the fragmentation, and do not waste it as ground vibration or noise.

Table 1. Average and maximum charge weight per delay for coal and siltstone blasts.

Initiation Sequence Type	Coal		Siltstone	
	Average	Maximum	Average	Maximum
Serpentine	17.9	26.5	14.7	18.8
30x50 ms parallel	28.6	59.5	21.6	56.1
30x60 ms parallel	75.9	148.8	88.8	224
25x42 ms parallel	74	74	59	115
30x105 ms parallel	36.7	93.4	38	78.2
30x30 ms diagonal	113.3	318.2	114.9	213.4
30x105 ms diagonal	35.9	134.2	40.8	57.5

Peak airblast values are shown in Figure 6 and plotted against the cube root scaled distance (CRSD). The cube root is used in the analysis of air pressure values because it correlates best with the cube-root of the charge weight (Konia, 1995; Persson, 1994). Air pressure intensities are attenuated when initiation sequence is changed and it varies with the different delay combinations. The diagonal method shows the lowest values for airblast just as it did for ground motions confirming that a more controlled and more efficient blast results in low vibrations and noise levels.

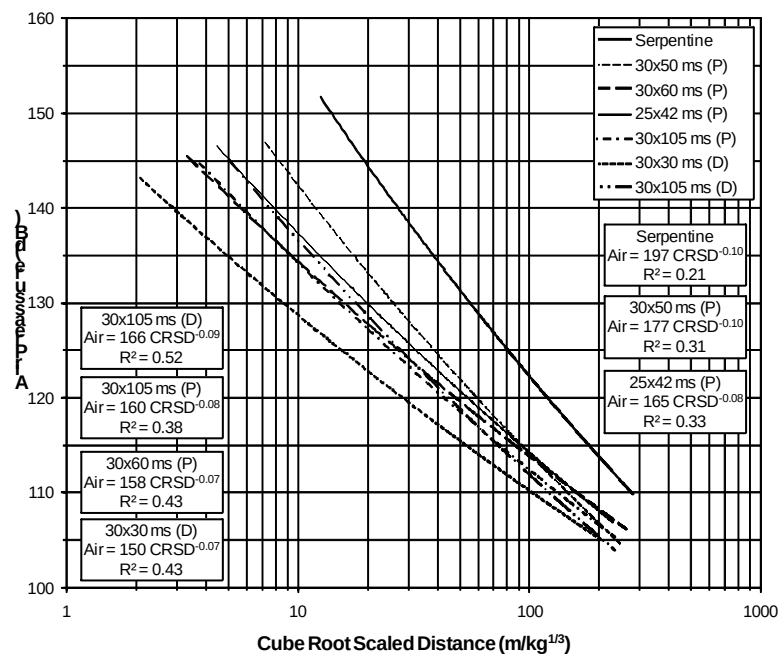


Figure 6. Air pressure (noise) versus cube root scaled distance attenuation plot.

The results from the initial changes in initiation sequence modifications of burden and spacing were suggested and evaluated. These changes resulted in a reduction of the number of holes drilled to obtain the same volume and implying cost reductions with the amount of explosives, accessories, meters drilled and labor. Hole depths were also increased, where two blasts were previously done to clear a thickness of siltstone or two adjacent coal seams with the serpentine initiation sequence, now one blast with deeper holes is required without increasing vibrations or resulting in poor fragmentation and effecting the excavation production.

Figures 7(a) and (b) are attenuation plots comparing the effects of changes in burden and spacing as well as different delay period combinations for a siltstone layer and coal blasting, respectively. In the siltstone analyzed, the initial burden and spacing, 3.7 x 4.3 meters (12 x 14 feet), show a low confinement and a low attenuation of ground motions. Increasing burden and spacing to 4 x 5 meters (13 x 16 feet) shows an increase in confinement and an increase in attenuation velocity, meaning that the high amplitudes will dissipate quicker.

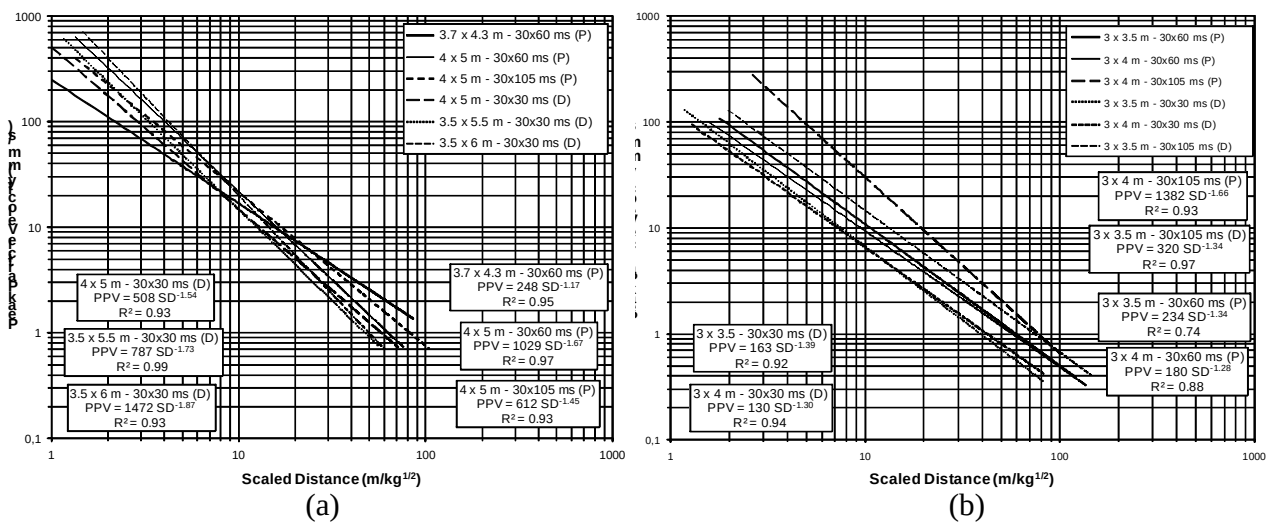


Figure 7. Attenuation plots comparing changes in blast design, (a) siltstone and (b) coal.

Using different delay combinations, the confinement is decreased and attenuation factor is kept high. Changing the initiation sequence from parallel to diagonal resulted in lower confinement with high attenuation. The last two comparisons show changes in burden and spacing using the diagonal initiation sequence of 30x30ms. Both changes show an increase in confinement as the spacing is increased. Attenuations of ground motions are high in both cases.

Burden and spacing used in coal shot patterns are tighter than in siltstone in order to control fragmentation. The siltstone in the overburden and interburdens are sent to the waste piles and fragmentation is not required to be small or more controlled. Blast must have sufficient energy to shock the rock and ease the excavation, not producing large boulders that will not fit in the excavators bucket

or the trucks. The more fragmented the coal is, by blasting, the lower will be the costs with crushing operations.

Figure 7(b) shows that burden and spacing variations in the coal shots do not have great influence in the confinement or attenuation of ground vibrations and delay combinations and initiation sequence have much more influence. A larger drill pattern will result in fewer explosives and accessories used as well as decrease the total meters drilled and labor. In addition, a tighter drill pattern will result in better fragmentation. There was a reduction in the percentage of coal product size 0 x 10 mm (0 x 0.39 in) and size over 38 mm (1,49 in) and an increase in the product size 10 x 38 mm (0.39 x 1.49 in) showing that better performed blasts resulted in a more uniform distribution of coal fragmentation fed to the crusher unit.

Finally, after conducting and analyzing the results obtained during the optimization process, the blast operation was standardized and the initiation sequence adopted was the diagonal with a 30 x 30 ms delay combination. Table 2 presents a summary of operation costs variation, per volume blasted and coal produced (“rom” run of mine). The blast optimization resulted in a 6.8 % cost savings per volume blasted in the operation even with an increase in explosive. Also, there was a 21.3 % cost reduction per ton of coal produced, even with a 24 % increase in production during the project.

Table 2. Summary of operation costs variations.

	Per volume blasted	Per ton rom
Explosives	+15.1%	-3.0%
Accessories	-4.4%	-20.5%
Drilling	-18.6%	-31.0%
Operation	-6.8%	-21.3%

Conclusion

With the introduction of the new approach for initiation sequence, which improved the degree of freedom, a better use of explosives energy was obtained. This became clear with the performance observed in excavator’s productivity by reducing siltstone and coal fragmentation and eliminating the need of re-drilling and blasting a bench that had been shot previously. Vibration and noise levels were lowered as shown by the attenuation equations resultant from the use of distinct combinations of delay periods in hole initiation and shot initiation sequence.

The optimization process for the diagonal initiation sequence method also shows a significant cost reduction. It was also possible to increase pattern and hole depth without influences in vibration and noise produced, even with the increase in charge weight per delay. The increase in hole depth resulted in less explosives, accessories and drilling costs as just one bench is necessary to clear the same amount of material compared to two benches in the “ serpentine” past design.

References

- Aimone-Martin, C.T., M.A. Martell, L.M. McKenna, D.E. Siskind, and C.H. Dowding, 2003, "Comparative Study of Structure Response to Coal Mine Blasting", prepared for the Office of Surface Mining Reclamation and Enforcement, Pittsburgh, USA.
- Dowding, C. H., 1985, "Blast Vibration Monitoring and Control", Prentice-Hall, Inc., Englewood Cliffs, NJ, USA.
- Konia, C.J., 1995, "Blasting Design". Intercontinental Development Corporation, Montville, Ohio, USA.
- Munaretti, E., 2002, "Desenvolvimento e Avaliação de Desempenho de Misturas Explosivas a Base de Nitrato de Amônia e Óleo Combustível", PhD Dissertation, Universidade Federal do Rio Grande do Sul, Brazil.
- Person, P., Holmberg, R. & Lee, J., 1994, "Rock Blasting and Explosives Engineering", CRC Press, London, England.
- Rosenhaim, V.L., 2005, "Response of a Residential Structure and Buried Pipelines to Construction Blasting in Basalt on the West Side of Albuquerque – NM", M.Sc. Thesis, New Mexico Institute of Mining and Technology.